Accelerated Math II/III Name_____

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HW #2 - Function Operations (Composition)

Date_____ Period____

Perform the indicated operation.

1) $g(a) = 2a + 4$
$h(a) = a^3 - 2a^2$
Find $(g \circ h)(a)$

2) $g(a) = -3a^2 - 4a$
Find $g(g(a))$

3) $g(a) = a + 5$
$f(a) = a^2 + 5a$
Find $(g \circ f)(a)$

4) $g(x) = 3x$
$h(x) = x - 4$
Find $(g \circ h)(x)$

5) $g(a) = a + 1$
$h(a) = a^2 + a$
Find $g(h(7))$

6) $g(n) = 4n - 2$
$f(n) = -n^2 - 2$
Find $(g \circ f)(6)$

7) $g(x) = -4x - 4$
Find $(g \circ g)(10)$

8) $f(x) = x - 1$
$g(x) = 4x + 5$
Find $f(g(8))$

9) $h(n) = 4n - 5$
Find $h(h(4n))$

10) $g(t) = 3t + 2$
$f(t) = t^2 + 3t$
Find $(g \circ f)\left(\frac{b}{4}\right)$

11) $g(t) = -t + 4$
$f(t) = t^2 + t$
Find $g(f(t - 2))$

12) $g(x) = x - 3$
$h(x) = -x^3 - 4x$
Find $(g \circ h)(x - 4)$

13) $h(t) = t - 1$
$g(t) = -4t - 2$
Find $2h(t) + 4g(t)$

14) $f(n) = 2n + 4$
$g(n) = n^3 + 3n$
Find $4f(n) + g(n)$

15) $f(x) = x^3 + 3x$
$g(x) = 3x - 2$
Find $4f(x) - 5g(x)$

16) $h(a) = 4a + 1$
$g(a) = 4a + 4$
Find $(4h + g)(a)$

17) $h(x) = x^3 + 3x^2$
$g(x) = 3x - 5$
Find $(2h - 2g)(-2)$

18) $h(n) = n^2 + 5$
$g(n) = 4n - 3$
Find $(5h - g)(3)$

19) $f(n) = -n - 1$
$g(n) = -2n^2 - 1$
Find $4f(8) - g(8)$

20) $f(x) = 2x - 3$
$g(x) = x^2 - 2x$
Find $3f(4) + g(4)$

21) $g(a) = 2a^2 - 3$
$h(a) = a - 1$
Find $-g(-3a) - 4h(-3a)$

22) $h(n) = n^2 - 3$
$g(n) = 3n + 3$
Find $3h(4n) - 5g(4n)$

23) $g(n) = 2n^3 - 5$
$f(n) = 4n$
Find $(g + 3f)(-2n)$

24) $h(a) = 4a - 5$
$g(a) = a^2 + 4a$
Find $(h - 2g)(2a)$

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Answers to HW #2 - Function Operations (Composition)

1) $2a^3 - 4a^2 + 4$
2) $-27a^4 - 72a^3 - 36a^2 + 16a$
3) $a^2 + 5a + 5$
4) $3x - 12$
5) 57
6) -154
7) 172
8) 36
9) $64n - 25$
10) $\frac{32 + 3b^2 + 36b}{16}$
11) $-t^2 + 3t + 2$
12) $-x^3 + 12x^2 - 52x + 77$
13) $-14t - 10$
14) $n^3 + 11n + 16$
15) $4x^3 - 3x + 10$
16) $20a + 8$
17) 30
18) 61
19) 93
20) 23
21) $-18a^2 + 12a + 7$
22) $48n^2 - 60n - 24$
23) $-16n^3 - 24n - 5$
24) $-8a^2 - 8a - 5$

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